ECSE 4320 Advanced Computer Systems

Project #1: SIMD Advantage Profiling

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Rensselaer Polytechnic Institute
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GitHub Repository: https://github.com/YOUR-USERNAME/YOUR-REPOSITORY

1 Introduction

[Write introduction here.]

2 Experimental Platform

All experiments were performed on the system described in Table 1.

Table 1: Experimental system configuration.

Component	Configuration
CPU model	AMD Ryzen 7 260
Physical cores	8
Threads	16
Logical CPUs / SMT	16
SIMD ISA support	SSE, AVX, AVX2, AVX-512
L1 cache	32 KB per core
L2 cache	1 MB per core
LLC / L3 cache	16 MiB shared
RAM	30 GB
Operating system	Linux Mint 22.3
Kernel version	7.0.0-38-generic
Compiler	GCC 13.3.0
Python	3.12.9
CPU frequency policy/governor	powersave

3 Benchmark Methodology

3.1 Selected Kernels

The following three or more computational kernels were evaluated:

Kernel 1

Kernel 2

Kernel 3

3.2 Correctness Validation

[Describe correctness procedure.]

3.3 Timing Method

[Describe timing methodology.]

3.4 Compiler Configurations

Table 2 summarizes the compiler configurations used for comparison.

Table 2: Compiler configurations.

Configuration	Compiler Flags	Purpose
Scalar-only	[Fill in]	Disable vectorization
Auto-vectorized	[Fill in]	Allow compiler vectorization
CPU-targeted	[Fill in]	Target available CPU features

4 Experiment 1: Scalar vs. SIMD

4.1 Question

[Write experimental question.]

4.2 Hypothesis

[Write pre-experiment hypothesis.]

4.3 Experimental Setup

[Describe setup.]

4.4 Results

The SIMD speedup is calculated as

$$S_{\text{SIMD}} = \frac{T_{\text{scalar}}}{T_{\text{SIMD}}}. \tag{1}$$

4.5 Analysis

[Analyze results.]

5 Experiment 2: Working-Set Size and Cache Locality

5.1 Question

[Write question.]

5.2 Hypothesis

[Write hypothesis before discussing results.]

5.3 Experimental Setup

[Describe setup.]

5.4 Results

x

5.5 Analysis

x

[Analyze results.]

6 Experiment 3: Alignment and Tail Handling

6.1 Question

[Write question.]

6.2 Hypothesis

[Write hypothesis.]

6.3 Experimental Setup

[Describe setup.]

6.4 Results

x

6.5 Analysis

x

[Analyze results.]

7 Experiment 4: Stride and Access Pattern

7.1 Question

[Write question.]

7.2 Hypothesis

[Write hypothesis.]

7.3 Experimental Setup

[Describe setup.]

7.4 Results

x

7.5 Analysis

[Analyze results.]

8 Experiment 5: Data Type

8.1 Question

[Write question.]

8.2 Hypothesis

[Write hypothesis.]

8.3 Experimental Setup

[Describe setup.]

8.4 Results

x

8.5 Analysis

[Analyze results.]

9 Vectorization Verification

9.1 Compiler Vectorization Report

```
[Insert a SHORT relevant compiler-report excerpt.]
```

[Explain what the compiler report demonstrates.]

9.2 Disassembly Evidence

```
[Insert only a few relevant assembly instructions.]
```

[Explain why this constitutes evidence of vector execution.]

10 Performance-Counter Analysis

Table 3: Selected performance-counter measurements.

Metric	Case A	Case B
Cycles	[Fill in]	[Fill in]
Instructions	[Fill in]	[Fill in]
Cache-related metric	[Fill in]	[Fill in]
Other relevant metric	[Fill in]	[Fill in]

[Interpret the counters rather than simply listing them.]

11 Roofline Analysis

11.1 Arithmetic Intensity

Arithmetic intensity is defined as

$$AI = \frac{\text{floating-point operations}}{\text{bytes transferred}}. \tag{2}$$

[Show your calculation.]

11.2 Machine Limits

The Roofline performance bound can be written as

$$P_{\text{attainable}} \leq \min\left(P_{\text{peak}}, AI \times B_{\text{memory}}\right), \tag{3}$$

where P_{peak} is the estimated peak floating-point throughput and B_{memory} is measured memory bandwidth.

[Describe peak-performance and bandwidth estimates.]

11.3 Interpretation

[Write Roofline interpretation.]

12 Negative / Failure Case

12.1 Hypothesis

[Write hypothesis.]

12.2 Results

[Present measurement/figure/table.]

12.3 Explanation

[Explain failure case.]

13 Experimental Limitations and Anomalies

[Discuss limitations and anomalies.]

14 AI Use Statement

[Write AI-use statement.]

15 Reproducibility

The complete source code, scripts, raw measurements, processed data, and plotting code are available in the project repository:

https://github.com/YOUR-USERNAME/YOUR-REPOSITORY

16 Conclusion

[Write conclusion.]